

Linear Algebra Notes: Vector Spaces, Subspaces, Gram-Schmidt & QR Decomposition

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1 Vector Space Axioms (IMG_0957)

A non-empty set V is called a **Vector Space** over a field $\mathbb{R}$ if $\forall u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$, the following axioms hold:

1. **Closure under Addition:** $u + v \in V$
2. **Closure under Scalar Multiplication:** $\alpha u \in V$
3. **Associativity of Addition:** $u + (v + w) = (u + v) + w$
4. **Additive Identity:** $u + 0 = u$
5. **Additive Inverse:** $u + (-u) = 0$
6. **Commutativity of Addition:** $u + v = v + u$
7. **Distributivity I:** $\alpha(u + v) = \alpha u + \alpha v$
8. **Distributivity II:** $(\alpha + \beta)u = \alpha u + \beta u$
9. **Compatibility of Scalar Multiplication:** $\alpha(\beta u) = (\alpha\beta)u = \beta(\alpha u)$
10. **Scalar Multiplication Identity:** $1 \cdot u = u$

Non-examples of Vector Spaces: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{I}$ (not closed under arbitrary real operations/lacks field closure)

Standard Examples of Vector Spaces: $\mathbb{R}, \mathbb{R}^n$, spaces of matrices $\mathbf{M}_{m \times n}$, polynomial spaces P_n

2 Linear Combinations, Span, Basis, and Dimension (IMG_0958 – IMG_0959)

Linear Combination

Let V be a vector space and $u_1, u_2, \dots, u_n \in V$. For scalars $\alpha_1, \alpha_2, \dots, \alpha_n$, the expression:

$$\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n$$

is called a **linear combination** of $u_1, u_2, \dots, u_n$.

Span of a Vector Space

A subset $B \subseteq V$ is said to **span** (or generate) V if any element $v \in V$ can be represented as a linear combination of elements of B .

Basis and Dimension

A subset $B \subseteq V$ is said to be a **basis** for V if:

1. B is **Linearly Independent (L.I.)**.
2. B **spans** V (i.e., $\text{span}(B) = V$).

The number of elements in a basis B is called the **dimension** of V , denoted $\dim(V)$.

Linear Independence Condition:

- If $\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n = 0 \implies \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$, then $\{u_1, \dots, u_n\}$ is **L.I.**
- If not all $\alpha_i = 0$, the set is **Linearly Dependent (L.D.)**.

Standard Basis Examples:

- Standard basis for $\mathbb{R}^2$: $\{(1, 0), (0, 1)\}$
- Standard basis for polynomials P_n : $\{1, x, x^2, \dots, x^n\}$
- Standard basis for 2×2 matrices: $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

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3 Solving Systems using Cramer's Rule (IMG_0959 – IMG_0960)

Solve the following linear system using Cramer's Rule:

$$\begin{cases} 3x + 7y + 8z = -13 \\ 2x + 0y + 9z = -5 \\ -4x + y - 26z = 2 \end{cases}$$

Determinants Calculation:

$$D = \begin{vmatrix} 3 & 7 & 8 \\ 2 & 0 & 9 \\ -4 & 1 & -26 \end{vmatrix} = 101$$

$$D_1 = \begin{vmatrix} -13 & 7 & 8 \\ -5 & 0 & 9 \\ 2 & 1 & -26 \end{vmatrix} = -707$$

$$D_2 = \begin{vmatrix} 3 & -13 & 8 \\ 2 & -5 & 9 \\ -4 & 2 & -26 \end{vmatrix} = 0$$

$$D_3 = \begin{vmatrix} 3 & 7 & -13 \\ 2 & 0 & -5 \\ -4 & 1 & 2 \end{vmatrix} = 101$$

Solution:

$$x = \frac{D_1}{D} = \frac{-707}{101} = -7, \quad y = \frac{D_2}{D} = \frac{0}{101} = 0, \quad z = \frac{D_3}{D} = \frac{101}{101} = 1$$

Final Solution: $(x, y, z) = (-7, 0, 1)$.

4 Four Fundamental Subspaces of a Matrix (IMG_0961 – IMG_0962)

For an $m \times n$ matrix A :

1. **Column Space $\mathcal{C}(A)$:** Spanned by the columns of A . Solutions to $AX = b$ exist iff $b \in \mathcal{C}(A)$.
2. **Null Space $\mathcal{N}(A)$:** The solution space of $AX = 0$.
3. **Row Space $\mathcal{C}(A^T)$:** Spanned by the rows of A .
4. **Left Null Space $\mathcal{N}(A^T)$:** The solution space of $A^T Y = 0$.

Example: Finding Vectors Orthogonal to Subspaces (IMG_0962)

Find a vector orthogonal to the row space and column space of:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

1. Vector Orthogonal to Row Space (Null Space $\mathcal{N}(A)$):

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

System of equations:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 0 \\ 2x_1 + 4x_2 + 3x_3 &= 0 \\ 3x_1 + 6x_2 + 4x_3 &= 0 \end{aligned}$$

Subtracting equations yields $x_3 = 0$, leading to $x_1 + 2x_2 = 0 \implies x_1 = -2x_2$. Setting $x_2 = 1$, the orthogonal vector is $[-2, 1, 0]^T$.

2. Vector Orthogonal to Column Space (Left Null Space $\mathcal{N}(A^T)$):

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

System of equations:

$$\begin{aligned} y_1 + 2y_2 + 3y_3 &= 0 \\ 2y_1 + 4y_2 + 6y_3 &= 0 \\ y_1 + 3y_2 + 4y_3 &= 0 \end{aligned}$$

Solving gives $y_2 = -y_3$ and $y_1 = y_2$. Setting $y_2 = 1$ gives the orthogonal vector $[1, 1, -1]^T$.

5 Projections and Gram-Schmidt Orthogonalization (IMG_0963 – IMG_0965)

Orthogonality and Projection

Two vectors u and v are **orthogonal** if their inner product is zero:

$$u \cdot v = 0$$

The **projection** of vector u onto vector v ($v \neq 0$) is defined as:

$$\text{Proj}_v(u) = \frac{u \cdot v}{\|v\|^2} v$$

Gram-Schmidt Orthogonalization Algorithm

Given a set of linearly independent vectors $\{u_1, u_2, \dots, u_n\}$, compute orthogonal vectors $\{v_1, v_2, \dots, v_n\}$:

$$\begin{aligned}v_1 &= u_1, & e_1 &= \frac{v_1}{\|v_1\|} \\v_2 &= u_2 - \frac{u_2 \cdot v_1}{\|v_1\|^2} v_1, & e_2 &= \frac{v_2}{\|v_2\|} \\v_3 &= u_3 - \frac{u_3 \cdot v_1}{\|v_1\|^2} v_1 - \frac{u_3 \cdot v_2}{\|v_2\|^2} v_2, & e_3 &= \frac{v_3}{\|v_3\|}\end{aligned}$$

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Gram-Schmidt Numerical Example (IMG_0964 – IMG_0965)

Find an orthogonal basis using Gram-Schmidt for the vectors:

$$u_1 = (1, 0, 1), \quad u_2 = (1, 1, 1), \quad u_3 = (1, 3, 4)$$

Step 1: Compute v_1

$$\begin{aligned}v_1 &= u_1 = (1, 0, 1) \\ \|v_1\|^2 &= 1^2 + 0^2 + 1^2 = 2\end{aligned}$$

Step 2: Compute v_2

$$\begin{aligned}u_2 \cdot v_1 &= (1, 1, 1) \cdot (1, 0, 1) = 1 + 0 + 1 = 2 \\ v_2 &= (1, 1, 1) - \frac{2}{2}(1, 0, 1) = (1, 1, 1) - (1, 0, 1) = (0, 1, 0) \\ \|v_2\|^2 &= 0^2 + 1^2 + 0^2 = 1\end{aligned}$$

Step 3: Compute v_3

$$\begin{aligned}u_3 \cdot v_1 &= (1, 3, 4) \cdot (1, 0, 1) = 1 + 0 + 4 = 5 \\ u_3 \cdot v_2 &= (1, 3, 4) \cdot (0, 1, 0) = 3 \\ v_3 &= (1, 3, 4) - \frac{5}{2}(1, 0, 1) - \frac{3}{1}(0, 1, 0) \\ v_3 &= (1, 3, 4) - \left(\frac{5}{2}, 0, \frac{5}{2}\right) - (0, 3, 0) = \left(-\frac{3}{2}, 0, \frac{3}{2}\right)\end{aligned}$$

Resulting Orthogonal Set:

$$\left\{ (1, 0, 1), (0, 1, 0), \left(-\frac{3}{2}, 0, \frac{3}{2}\right) \right\}$$

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6 QR Decomposition Setup (IMG_0966)

To find the QR decomposition of matrix $A = QR$:

- Q is an **orthogonal matrix** whose columns are orthonormal vectors derived via Gram-Schmidt ($e_1, e_2, \dots, e_n$).
- R is an **upper triangular matrix** consisting of inner products of original vectors with normalized basis elements.

Given matrix A formed by columns u_1, u_2, u_3 :

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The upper triangular matrix R is constructed as:

$$R = \begin{bmatrix} \|v_1\| & u_2 \cdot e_1 & u_3 \cdot e_1 \\ 0 & \|v_2\| & u_3 \cdot e_2 \\ 0 & 0 & \|v_3\| \end{bmatrix}$$

7 QR Decomposition Derivation & Example 1 (IMG_0967 – IMG_0968)

Theoretical Setup

Let $A = [a_1 \mid a_2 \mid \dots \mid a_n]$. Using the Gram-Schmidt process:

$$\begin{aligned} u_1 &= a_1, & e_1 &= \frac{u_1}{\|u_1\|} \\ u_2 &= a_2 - (a_2 \cdot e_1)e_1, & e_2 &= \frac{u_2}{\|u_2\|} \\ u_3 &= a_3 - (a_3 \cdot e_1)e_1 - (a_3 \cdot e_2)e_2, & e_3 &= \frac{u_3}{\|u_3\|} \end{aligned}$$

Then A can be decomposed into $A = QR$, where:

$$Q = [e_1 \mid e_2 \mid \dots \mid e_n] \quad (\text{Orthogonal matrix})$$

$$R = \begin{bmatrix} a_1 \cdot e_1 & a_2 \cdot e_1 & \dots & a_n \cdot e_1 \\ 0 & a_2 \cdot e_2 & \dots & a_n \cdot e_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_n \cdot e_n \end{bmatrix} \quad (\text{Upper Triangular matrix})$$

Example 1: Find the QR Decomposition of A

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 3 & 4 & 7 \end{bmatrix}$$

Column vectors: $a_1 = (1, 2, 3)$, $a_2 = (1, 3, 4)$, $a_3 = (2, 5, 7)$.

Step 1: Orthonormal Basis Vector e_1

$$u_1 = a_1 = (1, 2, 3), \quad \|u_1\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

$$e_1 = \frac{1}{\sqrt{14}}(1, 2, 3)$$

Step 2: Orthonormal Basis Vector e_2

$$a_2 \cdot e_1 = (1, 3, 4) \cdot \frac{1}{\sqrt{14}}(1, 2, 3) = \frac{1 + 6 + 12}{\sqrt{14}} = \frac{19}{\sqrt{14}}$$

$$u_2 = a_2 - (a_2 \cdot e_1)e_1 = (1, 3, 4) - \frac{19}{14}(1, 2, 3) = \left(-\frac{5}{14}, \frac{2}{7}, -\frac{1}{14}\right)$$

$$\|u_2\| = \sqrt{\frac{25 + 16 + 1}{196}} = \frac{\sqrt{42}}{14}$$

$$e_2 = \left(-\frac{5}{\sqrt{42}}, \frac{4}{\sqrt{42}}, -\frac{1}{\sqrt{42}}\right)$$

Step 3: Orthonormal Basis Vector e_3

$$u_3 = (2, 5, 7) - \left[(2, 5, 7) \cdot \frac{1}{\sqrt{14}}(1, 2, 3)\right] e_1 - [(2, 5, 7) \cdot e_2] e_2$$

$$u_3 = \left(\frac{11}{14}, 0, 0\right) \implies e_3 = (1, 0, 0)$$

Resulting Factorization Matrices:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{14}} & -\frac{5}{\sqrt{42}} & 1 \\ \frac{2}{\sqrt{14}} & \frac{4}{\sqrt{42}} & 0 \\ \frac{3}{\sqrt{14}} & -\frac{1}{\sqrt{42}} & 0 \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{14} & \frac{19}{\sqrt{14}} & \frac{33}{\sqrt{14}} \\ 0 & \frac{\sqrt{42}}{14} & a_3 \cdot e_2 \\ 0 & 0 & 0 \end{bmatrix}$$

8 QR Decomposition – Example 2 & Vector Orthogonal to Column Space (IMG_0969 – IMG_0970)

Example 2: Find QR Decomposition of A

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Steps:

$$u_1 = (1, 1, 0) \implies e_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$u_2 = (1, 0, 1) - \frac{1}{\sqrt{2}}e_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1 \right) \implies e_2 = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right)$$

$$u_3 = (0, 1, 1) - \frac{1}{\sqrt{2}}e_1 - \frac{1}{\sqrt{6}}e_2 = \left(-\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right) \implies e_3 = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

Resulting Decomposition Matrices:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}, \quad R = \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}$$

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Vector Orthogonal to Column Space of A (IMG_0970)

Find a vector orthogonal to the column space of $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$.

Solution: The orthogonal complement of $\mathcal{C}(A)$ is the left nullspace $\mathcal{N}(A^T)$:

$$A^T X = 0 \implies \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving the system yields $x_1 = x_3$ and $x_2 = -2x_3$. Setting $x_3 = 1$:

$$\mathbf{N}(A^T) = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

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9 Eigenvalues and Eigenvectors (IMG_0970 – IMG_0972)

Definition

For a square matrix A , a scalar λ and non-zero vector x are called an ****eigenvalue**** and ****eigenvector**** if:

$$Ax = \lambda x \implies (A - \lambda I)x = 0$$

Non-trivial solutions exist iff the characteristic determinant vanishes:

$$\det(A - \lambda I) = 0$$

Properties:

- $\sum \lambda_i = \text{Trace}(A)$
 - $\prod \lambda_i = \det(A)$
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Example 1: 2×2 Matrix (IMG_0971 – IMG_0972)

Find eigenvalues and eigenvectors for:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

Characteristic Equation:

$$\begin{vmatrix} 1 - \lambda & 2 \\ 1 & -\lambda \end{vmatrix} = 0 \implies \lambda^2 - \lambda - 2 = 0 \implies (\lambda - 2)(\lambda + 1) = 0$$

Eigenvalues: $\lambda_1 = 2, \lambda_2 = -1$.

Eigenvector for $\lambda = 2$:

$$\begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -x_1 + 2x_2 = 0 \implies x_1 = 2x_2 \implies v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Eigenvector for $\lambda = -1$:

$$\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x_1 + x_2 = 0 \implies x_1 = -x_2 \implies v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

10 Matrix Diagonalization & 3×3 Example (IMG_0973 – IMG_0976)

Diagonalization Theorem

If matrix A has n linearly independent eigenvectors, it can be diagonalized as:

$$D = X^{-1}AX$$

where $X = [v_1 \mid v_2 \mid \cdots \mid v_n]$ is the eigenvector matrix, and $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$.

Example: Diagonalizing a 3×3 Matrix (IMG_0973 – IMG_0976)

Find the diagonalized matrix for:

$$A = \begin{bmatrix} 5 & 10 & -10 \\ 10 & 5 & -20 \\ 5 & -5 & -10 \end{bmatrix}$$

Step 1: Eigenvalues Expanding $|A - \lambda I| = 0$:

$$\lambda(\lambda + 15)(\lambda - 15) = 0 \implies \lambda_1 = 0, \lambda_2 = -15, \lambda_3 = 15$$

Step 2: Eigenvectors

- For $\lambda = 0$: $v_1 = [2, 0, 1]^T$
- For $\lambda = -15$: $v_2 = [0, 1, 1]^T$
- For $\lambda = 15$: $v_3 = [1, 1, 0]^T$

Step 3: Transformation Matrix X and Verification

$$X = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad X^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 2 \\ 1 & 2 & -2 \end{bmatrix}$$

$$D = X^{-1}AX = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -15 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

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11 Quadratic Forms Identification (IMG_0976)

Identify the geometric shape represented by the quadratic form:

$$17x_1^2 - 30x_1x_2 + 17x_2^2 = 128$$

Matrix Representation:

$$A = \begin{bmatrix} 17 & -15 \\ -15 & 17 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Eigenvalues of Matrix A :

$$\det(A - \lambda I) = (17 - \lambda)^2 - 225 = 0 \implies 17 - \lambda = \pm 15$$

$$\lambda_1 = 2, \quad \lambda_2 = 32$$

Canonical Standard Form (Ellipse): Using the spectral transformation $y_1^2\lambda_1 + y_2^2\lambda_2 = 128$:

$$2y_1^2 + 32y_2^2 = 128 \implies \frac{y_1^2}{64} + \frac{y_2^2}{4} = 1 \implies \frac{y_1^2}{8^2} + \frac{y_2^2}{2^2} = 1$$

Conclusion: The quadratic equation represents an **ellipse** with semi-major axis $a = 8$ and semi-minor axis $b = 2$.

12 Complex Matrices (IMG_0977 – IMG_0979)

Definitions

Let $\bar{A}^T = A^H$ denote the conjugate transpose (Hermitian transpose) of a complex matrix A :

- **Hermitian Matrix:** $\bar{A}^T = A$
- **Skew-Hermitian Matrix:** $-\bar{A}^T = A$ (or $\bar{A}^T = -A$)
- **Unitary Matrix:** $A\bar{A}^T = I \implies \bar{A}^T = A^{-1}$

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Example 1: Eigenvalues & Eigenvectors of a Hermitian Matrix

Let $A = \begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$. A is Hermitian since $\bar{A}^T = A$.

Characteristic Equation & Eigenvalues:

$$|A - \lambda I| = 0 \implies \begin{vmatrix} 4-\lambda & 1-3i \\ 1+3i & 7-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)(7-\lambda) - (1-3i)(1+3i) = 0$$

$$\lambda^2 - 11\lambda + 28 - (1+9) = 0 \implies \lambda^2 - 11\lambda + 18 = 0$$

$$\lambda^2 - 9\lambda - 2\lambda + 18 = 0 \implies (\lambda-9)(\lambda-2) = 0$$

$$\lambda_1 = 9, \quad \lambda_2 = 2 \quad (\text{Both Real})$$

Eigenvector for $\lambda = 9$:

$$(A - 9I)X = 0 \implies \begin{bmatrix} -5 & 1-3i \\ 1+3i & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-5x_1 + (1-3i)x_2 = 0 \implies x_1 = \left(\frac{1-3i}{5}\right)x_2$$

Setting $x_2 = 5$:

$$X_1 = \begin{bmatrix} 1-3i \\ 5 \end{bmatrix}$$

Eigenvector for $\lambda = 2$: Similarly, for $\lambda = 2$:

$$X_2 = \begin{bmatrix} -1+3i \\ 2 \end{bmatrix}$$

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Theorems on Complex Matrices (IMG_0978)

- (a) The eigenvalues of a Hermitian (or real symmetric) matrix are all **real**.
- (b) The eigenvalues of a Skew-Hermitian (or real skew-symmetric) matrix are either **purely imaginary or zero**.
- (c) The eigenvalues of a Unitary (or real orthogonal) matrix have an **absolute value of 1** ($|\lambda| = 1$).

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Orthonormal Verification (IMG_0978 – IMG_0979)

For normalized eigenvectors matrix $X = [x_1 \mid x_2]$:

$$\bar{x}_1^T x_1 = 1, \quad \bar{x}_2^T x_2 = 1, \quad \bar{x}_1^T x_2 = 0, \quad \bar{x}_2^T x_1 = 0$$

Verification Computations:

$$x_1 = \begin{bmatrix} \frac{1+i}{\sqrt{2}} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, \quad x_2 = \begin{bmatrix} -\frac{1+i}{\sqrt{2}} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\bar{x}_1^T x_1 = \frac{1-i^2}{4} + \frac{2}{4} = \frac{1}{2} + \frac{1}{2} = 1$$

$$\bar{x}_2^T x_2 = \frac{1^2 - i^2}{4} + \frac{1}{2} = \frac{2}{4} + \frac{1}{2} = 1$$

$$\bar{x}_1^T x_2 = \left(\frac{1-i}{2} \cdot \frac{\sqrt{2}}{2} \right) \left[-(1+i), \sqrt{2} \right]^T = -\frac{(1+i)}{4} + \frac{2}{4} = 0$$

$$\bar{x}_2^T x_1 = -\frac{(1-i)}{2} \cdot \frac{\sqrt{2}}{2} \left(\frac{1+i}{2}, \frac{\sqrt{2}}{2} \right) = 0$$

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Hermitian Forms (IMG_0979)

If A is a Hermitian matrix and X is a vector, then:

$$\bar{X}^T A X \quad \text{is called a **Hermitian Form**.}$$

(Note: If A is Skew-Hermitian, then $\bar{X}^T A X$ is called a Skew-Hermitian Form).

Example: Find the Hermitian form of A and X

$$A = \begin{bmatrix} 3 & 2-i \\ 2+i & 4 \end{bmatrix}, \quad X = \begin{bmatrix} 1+i \\ 2i \end{bmatrix}$$

$$\bar{X}^T = [1-i \quad -2i]$$

$$\begin{aligned} \bar{X}^T A X &= [1-i \quad -2i] \begin{bmatrix} 3 & 2-i \\ 2+i & 4 \end{bmatrix} \begin{bmatrix} 1+i \\ 2i \end{bmatrix} \\ &= [1-i \quad -2i] \begin{bmatrix} 3+3i+4i+2 \\ 2+i+3i+8i \end{bmatrix} = [1-i \quad -2i] \begin{bmatrix} 5+7i \\ 2+11i \end{bmatrix} \\ &= (5+7i-5i+7) - (4i-22) = 5+7+22 = \mathbf{34} \quad (\text{Real}) \end{aligned}$$

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13 Similar Matrices (IMG_0980 – IMG_0982)

Definition & Properties

An $n \times n$ matrix $\hat{A}$ is called ****similar**** to an $n \times n$ matrix A if:

$$\hat{A} = P^{-1}AP$$

for some non-singular matrix P .

Key Properties: If A and $\hat{A} = P^{-1}AP$ are similar, they share:

1. Same Rank
2. Same Determinant
3. Same Trace
4. Same Eigenvalues

Theorem on Eigenvectors: If Y is an eigenvector of $\hat{A}$, then the corresponding eigenvector of A is:

$$X = PY$$

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Example 1: Eigenvalues and Eigenvectors of Similar Matrices (IMG_0981 – IMG_0982)

Given:

$$A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}, \quad P = \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$$

$$P^{-1} = -\frac{1}{2} \begin{bmatrix} -1 & -2 \\ -3 & -4 \end{bmatrix}$$

Step 1: Compute Similar Matrix $\hat{A} = P^{-1}AP$

$$\begin{aligned}\hat{A} &= -\frac{1}{2} \begin{bmatrix} -1 & -2 \\ -3 & -4 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} \\ &= -\frac{1}{2} \begin{bmatrix} -1 & -2 \\ -3 & -4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -25 & 11 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 50 & -24 \\ 100 & -50 \end{bmatrix} = \begin{bmatrix} -25 & 12 \\ -50 & 25 \end{bmatrix}\end{aligned}$$

Step 2: Eigenvalues of A

$$\begin{aligned}|A - \lambda I| &= \begin{vmatrix} 3 - \lambda & 4 \\ 4 & -3 - \lambda \end{vmatrix} = 0 \implies \lambda^2 - 9 - 16 = 0 \implies \lambda^2 - 25 = 0 \\ \lambda &= \mathbf{5}, \quad \mathbf{-5}\end{aligned}$$

Step 3: Eigenvalues of $\hat{A}$

$$\begin{aligned}|\hat{A} - \lambda I| &= \begin{vmatrix} -25 - \lambda & 12 \\ -50 & 25 - \lambda \end{vmatrix} = 0 \implies \lambda^2 - 625 + 600 = 0 \implies \lambda^2 - 25 = 0 \\ \lambda &= \mathbf{-5}, \quad \mathbf{5}\end{aligned}$$

(Confirms that A and $\hat{A}$ have identical eigenvalues).

Step 4: Eigenvector Transformation Relation ($X = PY$)

- For $\lambda_1 = 5$: Eigenvector of A is $X = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, Eigenvector of $\hat{A}$ is $Y = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

$$PY = \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = X$$

- For $\lambda_2 = -5$: Eigenvector of A is $X = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, Eigenvector of $\hat{A}$ is $Y = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.

$$PY = \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} \equiv -2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} = X$$